

ADOS-2 | Autism Diagnostic Observation Schedule, Second Edition

Lord, Rutter, DiLavore, Risi, Gotham, & Bishop (2012)
Lord, Luyster, Gotham, & Guthrie (2012)



ADOS[®]-2 Training for Clinicians

Philadelphia, PA / Day 1: May 1, 2025
A Live Event by The WPS Professional Learning Center (ProLearn[®])

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Goals of This Training

- Introduction to ADOS-2, including why the ADOS-2 was developed
- Demonstration of ADOS-2 to illustrate administration and coding
- Introduction to coding conventions
- Discussion of appropriate use and interpretation
- Information necessary for further use

This training presumes background in diagnostic evaluation of neurodevelopmental disorders generally, as well as understanding of autism spectrum disorders (ASD) specifically.

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What This Training Is Not

- Equivalent to full preparation for clinical or research use
- A replacement for reading the Manual!
- Training and experience in diagnostic assessment of autism spectrum disorder (ASD)

Further practice and experience will be needed before you can use the ADOS-2 for clinical or research purposes.

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Day 1: Today's Agenda

- Orientation to DSM-5-TR ASD Criteria
- Introduction to ADOS-2
- Introduction to Modules 3 and 4
- Demonstration of Module 3 or 4
- Independent coding
- Review administration
- Scoring the protocol and algorithm
- Calculating reliability
- Remaining questions about Modules 3 and 4

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Diagnostic Evaluation of Autism Spectrum Disorder (ASD)

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DSM-5-TR: Autism Spectrum Disorder

- A. Persistent deficits in social communication and social interaction across contexts, as manifested by all of the following, currently **or by history**,
1. Deficits in social-emotional reciprocity....
 2. Deficits in nonverbal communicative behaviors used for social interaction....
 3. Deficits in developing, maintaining, and understanding relationships....



Source: Autism Spectrum Disorder, in *DSM-5-TR* (pp. 56–58), 2022.

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DSM-5-TR: Autism Spectrum Disorder

(cont.)

- B. Restricted, repetitive patterns of behavior, interests, or activities as manifested by at least two of the following, currently **or by history**,
1. Stereotyped or repetitive motor movements, use of objects, or speech...
 2. Insistence on sameness, inflexible adherence to routines, or ritualized patterns of verbal or nonverbal behavior...
 3. Highly restricted, fixated interests that are abnormal in intensity or focus...
 4. Hyper- or hypo-reactivity to sensory input or unusual interest in sensory aspects of environment...

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DSM-5-TR: Autism Spectrum Disorder

(cont.)

- C. Symptoms must be present in early developmental period (but may not become fully manifest until social demands exceed limited capacities or may be masked by learned strategies in later life)
- D. Symptoms cause clinically significant impairment in ... current functioning
- E. Disturbances not better explained by ID, DD



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Heterogeneity Is a Hallmark of ASD

- Age, developmental level (e.g., IQ, mental age), expressive language level significantly affect how symptoms manifest.
- Also critical to consider biological sex, gender identity, ethnicity, culture, and context (e.g., different settings or social circumstances) when evaluating social–communication strengths and challenges.

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Specifiers

Meant to indicate possible etiological factors that could impact course	Descriptive, meant to indicate another important factor	Severity specifiers indicate support needs
- Associated with medical/genetic diagnoses (e.g., ASD with fragile X syndrome) - Associated with neurodevelopmental, mental health, behavioral disorder (e.g., ASD with ADHD or ASD with ID)	- With or without intellectual impairment (e.g., ASD with intellectual disabilities) - Advise verbal and nonverbal due to unevenness - With or without structural language impairment with level of language (e.g., ASD with phrase speech) - With catatonia (e.g., ASD with catatonia)	- Social communication - Requires support - Requires substantial support - Requires very substantial support - Repetitive behaviors - Requires support - Requires substantial support - Requires very substantial support

Source: DSM-5-TR, 2022, p. 57.

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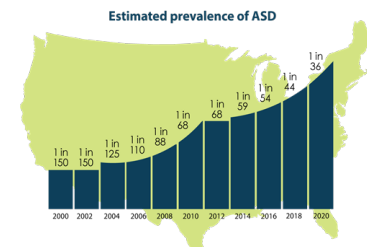
Increasing Prevalence Over Time

1 in 36 in study year (SY) 2020 (1 in 23 boys and 1 in 88 girls)

In 2020–2021, ASD category represented 12% of children in special education in the U.S. (NCES, 2022)

Important caveats:

- Prevalence differs by state and by community.
- Prevalence results differ based on methods used.
- Much of the increase in prevalence over time is due to expanded diagnostic criteria and improved detection.
 - In 2018–2020, increase in prevalence was accompanied by increases in identification of Hispanic, Asian, and Black children.



Source: Maenner et al., 2023.

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Changes in Epidemiology

- Greatest increase in number of new cases of ASD is accounted for by individuals with average or above average IQ
- Historically, 70%–85% of autism cases were associated with intellectual disability, but this is no longer the case
 - 14 recent population-based studies reported IQ data: across samples, 30% to 85.3% had IQ in the average range
- It is now estimated that 75%–90% of children with ASD acquire some functional language (compared to earlier estimate of 50%)

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Demographic Considerations in Diagnosis

- Sex: Male-to-female ratio has been consistently reported at approximately 4:1 (e.g., Zeidan et al., 2022)
 - Ratio changes depending on level of IQ (more females with lower IQ, more identified males with higher IQ)
 - Some females at risk for being identified later
 - Studies of sex differences are hugely hindered by cohort effects and ascertainment biases

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Demographic Considerations in Diagnosis (cont.)



- Ethnicity/Cultural Differences:
 - BIPOC children in the U.S. are identified later (e.g., Constantino et al., 2020)
 - Professionals are more likely to miss ASD symptoms in children from some cultural and socioeconomic groups
 - Families may have different expectations around social interaction or play behaviors

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Important Elements of an ASD Diagnostic Assessment

- Medical evaluation and hearing screening
- Interview with parents/caregivers
 - Developmental history
 - Pervasiveness of behaviors
 - History of restricted and repetitive behaviors
- Child assessment
 - Direct observation of current ASD-related symptoms and behaviors
 - Cognitive and language testing
- Observation in and/or reports about behavior in relevant environments
 - At home, school, daycare

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What a Comprehensive Assessment Should Identify

- Strengths and weaknesses in developmental skills
 - What can we capitalize on?
 - Where do we need to provide extra support?
- Are there notable challenges in certain areas of development?
- How is the child functioning with everyday tasks?
 - Adaptive skills often much lower than expected based on IQ
- What are their receptive and expressive language skills?
- How do they use their language skills spontaneously, across pragmatic contexts?

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Developmental History Is Important to Consider

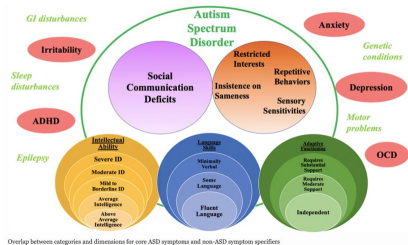
- DSM-5: Difficulties must be present during the early developmental period
 - Even with later referral, it is important to establish history of early difficulties in social–communication
- Onset patterns vary significantly and represent another example of heterogeneity
 - “Regressive” onset thought to occur in approximately 20%–30% of cases (Ozonoff et al., 2015; Thurm et al., 2017)
 - Rather than frank loss, many children may “plateau” or fail to progress in skill acquisition
 - Some children may also show abnormalities prior to plateauing or loss

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ASD or Not Is Only the First Question

- Assessing symptoms in non-core domains is not only important for differential diagnosis but also for determining comorbid diagnoses (e.g., ASD and ADHD, ASD and depressive disorder)



Source: Rosen, N. E., Lord, C., & Volkmar, F. R. (2021). The diagnosis of autism: From Kanner to DSM-III to DSM-5 and beyond. *Journal of Autism and Developmental Disorders*, 4253–4270.

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Next Question: Where Are They on the Spectrum?

- DSM-5 severity indicators and other quick indicators of “severity” may not be clinically useful or appropriate.
- It is likely more meaningful to describe an individual’s particular profile AT A MOMENT IN TIME:
 - How old are they?
 - What is their cognitive ability? Language ability? Adaptive functioning? Academic functioning?
 - How do their social–communication abilities compare to other children of a similar age and developmental level?
 - How do their restrictive and repetitive behaviors (RRBs) compare to other children of a similar age and developmental level?
 - Do they have internalizing and/or externalizing behavior problems?

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Summary

- Diagnostic assessment of ASD (or anything else) requires a lot of training, experience, and willingness to admit the limits of what we know.
- Best practices for ASD diagnostic assessment require evaluation across multiple domains of functioning and across settings.
- For direct observation, the ADOS-2 is ONE tool that may be helpful in obtaining information relevant to a diagnosis of ASD.

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Why was the ADOS developed?

What We Observe Is Affected by the Context We Create

- The ADOS is intended to provide an opportunity to elicit and observe behaviors associated with ASD.
- It was developed to facilitate standardization of observation across different children, different clinicians, and different sites.
- The ADOS attempts to address several challenges inherent to the assessment of behaviorally defined conditions.

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Response to Diagnostic Challenges

- **Challenge:** Variability in how direct observations are conducted (setting, clinician behavior, materials, demands on participant)



- **Solution:** The ADOS-2 provides a means of standardizing direct observation to promote consistency across clinicians and sites

ADOS-2 is standardized by:

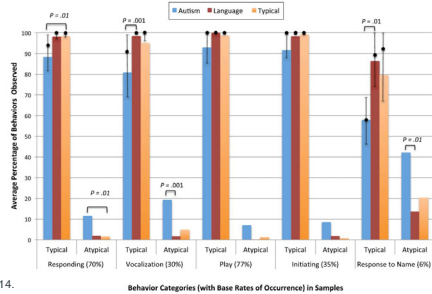
- ✓ Materials
- ✓ Behavior of the examiner
- ✓ Behaviors to be observed
- ✓ How the individual's behaviors are quantified
- ✓ Training of examiner



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Pitfalls in Relying on Brief Observations

- Children with ASD showed “typical” behaviors 89% of the time
- Expert reviewers missed 39% of ASD cases



Source: Gabrielson et al., 2014.

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Response to Diagnostic Challenges (cont.)

- Challenge:** Variability in defining ASD symptoms and behaviors
 - Because the DSM is just a guide, there can be variability in how people interpret the diagnostic criteria of ASD.
- Solution:** The ADOS-2 provides operationalized, standardized criteria for defining behaviors associated with ASD.

Focusing on Specific Abnormalities

- Detailed description of behaviors
- Operationalized criteria
- Focus on key situations
- Comparison to control group

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Response to Diagnostic Challenges (cont.)

- Challenge:** Developmental changes in autism
 - Expressive language level is an ability that affects almost every aspect of social interaction and play
- Solution:** Module system (different “presses” and codes by age/language level)
 - Minimizes the effect of language level/age (five modules)

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ADOS-2 General Administration Guidelines

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The ADOS-2 Is a Clinical Instrument

- The ADOS-2 helps a clinician create a “social world” in which ASD-related behaviors can be observed.
- It helps construct a profile of strengths and difficulties that may be useful in intervention planning.
- Ultimately, the quality of the information derived from the ADOS-2 is dependent on the examiner’s experience and sensitivity (to act and not to act).
- The ADOS-2 is to be used as just *one* source of information: no single instrument, score, or classification is ever sufficient to justify a diagnosis.

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ADOS-2 Design

- Takes approximately 40–60 minutes to administer (additional time is needed to score after administration)
- Based on the participant’s **expressive language abilities (and age)**, the examiner selects and administers *one* of the Modules
- Structured and unstructured activities, which vary depending on the Module, are used to elicit ASD-related symptoms and behaviors
- Individual behaviors are coded, and an algorithm is used to determine diagnostic classification
 - Diagnostic classification will not always match the overall clinical diagnosis

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The ADOS-2 Is NOT Appropriate for Everyone

- Designed for individuals aged 12 months to adulthood AND with a minimum nonverbal mental age of at least 12 months (for Toddler Module) and 15–18 months (Module 1)
- Not validated for individuals with significant sensory impairments (e.g., blindness, deafness)
- Not validated for individuals with significant motor impairments
- As with any test, scores may be invalid in other situations (e.g., when using masks or other PPE)

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Guidelines for Selecting a Module

Language **FIRST** and **THEN** age

(see ADOS-2 Manual, p. 10)

Module	Expressive Language Level Requirements	Chronological Age Recommendations
T	No speech to simple phrases	12 to 30 months
1	No speech to simple phrases	At least 31 months; may not be appropriate past school age
2	Three-word phrases/ not yet verbally fluent	May not be appropriate past school age
3	Regular use of complex sentences	Child or younger adolescent (≈16 and younger)
4	Regular use of complex sentences	Older adolescent or adult

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Selecting Module 1

Module	Expressive Language Level Requirements	Chronological Age Recommendations
T	No speech to simple phrases	12 to 30 months
1	No speech to simple phrases	At least 31 months; may not be appropriate past school age
2	Three-word phrases/ not yet verbally fluent	May not be appropriate past school age
3	Regular use of complex sentences	Child or younger adolescent (≈16 and younger)
4	Regular use of complex sentences	Older adolescent or adult

- Nonverbal to primarily single words and occasional phrases
If flexible three-unit phrases → Module 2
 - Regardless of age!
- 31 months and older
If 30 months or younger → Toddler Module is recommended
Minor modifications may be useful for older children and adolescents
 - Or use of the ADOS-2 may not be appropriate in these cases

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Selecting Module 2

Module	Expressive Language Level Requirements	Chronological Age Recommendations
T	No speech to simple phrases	12 to 30 months
1	No speech to simple phrases	At least 31 months; may not be appropriate past school age
2	Three-word phrases/ not yet verbally fluent	May not be appropriate past school age
3	Regular use of complex sentences	Child or younger adolescent (≈16 and younger)
4	Regular use of complex sentences	Older adolescent or adult

- Flexible phrase speech, but not regular use of complex sentences
Phrase speech: regular, spontaneous, non-echoed utterances of at least three words, usually including a verb
Typical expressive language level of between 24 and 48 months
- Minor modifications may be useful for older children or adolescents
Or use of the ADOS-2 may not be appropriate in some cases

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Selecting Module 3

Module	Expressive Language Level Requirements	Chronological Age Recommendations
T	No speech to simple phrases	12 to 30 months
1	No speech to simple phrases	At least 31 months; may not be appropriate past school age
2	Three-word phrases/ not yet verbally fluent	May not be appropriate past school age
3	Regular use of complex sentences	Child or younger adolescent (≈16 and younger)
4	Regular use of complex sentences	Older adolescent or adult

- Regular use of complex sentences
 - Expressive language skills of a typical 4-year-old
 - Produces a range of sentence types and grammatical forms
 - Provides information beyond immediate context
 - Uses logical connections such as "but" and "because"
- Appropriate for children and most younger adolescents and sometimes for older adolescents and adults
 - Depending on developmental level and interest in toys
- Can modify language and tone depending on age of child

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Selecting Module 4

Module	Expressive Language Level Requirements	Chronological Age Recommendations
T	No speech to simple phrases	12 to 30 months
1	No speech to simple phrases	At least 31 months; may not be appropriate past school age
2	Three-word phrases/ not yet verbally fluent	May not be appropriate past school age
3	Regular use of complex sentences	Child or younger adolescent (≈16 and younger)
4	Regular use of complex sentences	Older adolescent or adult

- Regular use of complex sentences
Expressive language skills of a typical 4-year-old
Produces a range of sentence types and grammatical forms
Provides information beyond immediate context
Uses logical connections such as "but" and "because"
 - Putting two ideas together: "I like to go swimming because I like splashing my brother."
- Appropriate for most older adolescents and adults with some level of independence in everyday choices

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BEFORE the Administration

- Prepare room and materials *before* the participant arrives
- Arrange the environment
 - Choose appropriately sized table and chairs
 - Set the tasks out but not on the table you are working at; hide them (e.g., cover them with a blanket)
- Proper materials
 - Make sure your equipment works (e.g., bubble toy, remote car or bunny, balloons)
 - Have a stash of extras, including batteries
 - Have the proper snack materials (e.g., ask about food restrictions or allergies)

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BEFORE the Administration (cont.)

- Select what you think is the correct module, but have alternate protocol and materials ready in case you need to switch modules
- Know how to administer activities without the Manual
 - Read the Manual carefully and practice, practice, practice!

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Tips and Tricks: Before Administration

- Make a space at the top of each task to note eye contact and shared enjoyment
- Read through the codes right before you start administering
- Develop a shorthand notation that works for you

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Shorthand Notation

(see ADOS-2 Manual, p. 16)

Examiner: Ex.	Eye contact: EC or 2 dots (to represent a pair of eyes)
Facial expression: FE or DFE (if directed) plus description (e.g., DFE angry)	Spontaneous Initiation of joint attention: IJA
Point: Pt. or Pt. + EC (when coordinated with eye contact)	Descriptive gesture: DG plus description (e.g., DG flying, DG size)
Emotional/emphatic gestures: Emot. G or EG	Social overtone: SOV (with + or – to indicate the quality of the overtone)
Offers information: OI plus theme (e.g., OI videogames)	Asks for information: AI plus description of question
Immediate echolalia: echo	Self-injurious behavior: SIB
Hand mannerism: HM	Visual inspection: VI and item (e.g., VI disk)
Shared enjoyment: SE	Complex mannerism: CM or CBM

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DURING the Administration

- Take a language sample at the beginning of the administration to confirm you are using the correct module before proceeding too far
 - If you switch modules, *exclude* behaviors observed during tasks that are exclusive to the “incorrect” module when coding
- Move from less structure to more structure
- Behavior and response of child in part depends on examiner’s sensitivity to respond
 - Knowing when to act and when not to act
- Be flexible—with the materials, with the space, with yourself and your behavior
 - You CAN administer tasks in any order

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DURING the Administration (cont.)

- The ADOS was designed to be administered by the person making (or heavily involved in making) the diagnostic decisions.
 - The ADOS should not be “ordered” and then interpreted by someone else.
- There should only be *one* examiner.
 - *Examiner needs to take notes for themselves.*
 - Codes should be assigned only with regard to behavior directed to the *examiner* (and/or parent/caregiver as specified) and NOT the observer.
- For Modules T–2, make sure that a parent or familiar caregiver is present. Involve this person in the administration as specified in the Manual.
- If there is an observer, they can also take notes for consulting and consensus coding purposes, but the examiner’s codes should be based on their own observations and notes during the administration.

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AFTER the Administration

- Assign ADOS-2 codes *immediately* following the assessment
- ONLY code behavior that was observed DURING the ADOS-2
- Codes should be assigned based on live (not recorded) administration
 - Importantly, psychometric data are based on information from live coding
 - Video recordings can be used for detailed study, assessment of reliability, etc.
- Use information from the ADOS-2 in combination with all other information from the diagnostic assessment
 - Remember that the ADOS-2 Classification will not always match the overall clinical diagnosis for every child

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Discussion of General Coding Guidelines

See pages 16–22 of ADOS-2 Manual

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ADOS-2 Coding

- One of the most important ways that the ADOS-2 is standardized is in how the observations are used and coded
- Coding is based on the behavioral samples obtained during ADOS-2 activities
- Code **ONLY** what you see during the ADOS-2
 - Behaviors observed before or after the ADOS-2 can be incorporated into overall diagnostic impressions

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ADOS-2 Items

- There are five coding sections
 - Sections are coded based on developmental OR chronological age standards (headings will specify this)
 - Section guidelines are overruled by the item's specific coding guidelines to the contrary (e.g., Module 3, Item A8)
- Each item has particular coding instructions
 - Use coding conventions as a general guide, but attend to the specific instructions in each code
- Instructions may differ between modules even though the title of the code is the same
 - You must read the full paragraph above the code for each item

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ADOS-2 Items (cont.)



- Some items consider a single skill (e.g., Gestures, Eye Contact, Giving)
- Some items consider a particular activity (e.g., Response to Name)
- Some items are summary codes (e.g., Quality and Amount codes)
- Summary codes take into account behaviors already considered in other codes
 - Showing a toy or using the examiner's hand as a tool to request are types of social overtures
 - Answering questions with stereotyped phrases could affect the Quality of Social Response
- Majority of examples vs. the best example vs. most abnormal example

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Coding Conventions

- 0 = Abnormality of type specified is NOT present (not necessarily "normal")
- 1 = Abnormality of type specified is present, but not sufficiently clear, severe, frequent, or marked to receive a code of 2
- 2 = Abnormality of type specified is definitely present and meets specific mandatory criteria
- 3 = Abnormality of type specified is present to a degree that may interfere with functioning
- 7 = Definite abnormality in general area of coding, but not of type specified
- 8 = Not applicable
- 9 = Unknown/missing

For reliability, the goal is 80% exact item agreement on protocol and algorithm, NOT including 2 vs. 3 as disagreements.

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Other Coding Principles

- When between two codes, code conservatively (i.e., choose the less severe code)
- Do not code the same aspect of a behavior in 2 different codes (i.e., do not double code)
- **Code the *behavior* (what you see); *interpret it later***

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ADOS-2 Administration Modules 3 and 4

Modules 3 and 4 Administration

- Don't let it be too easy (don't be too nice or too structured)
- Keep at it; don't rush
- Make sure there is some time when it's fun as well
- Parent/caregiver not in the room
- Preferably sit at a 90-degree angle; large enough table for Make-Believe Play (Module 3)

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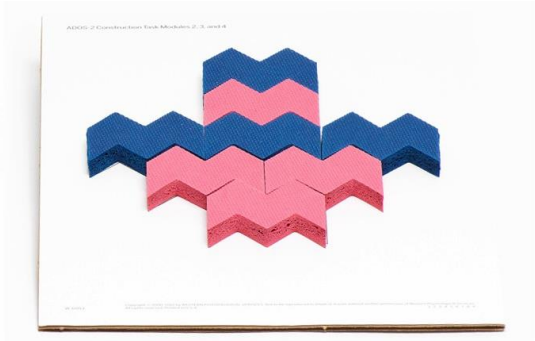
Module 3 Activities Fluent Speech

Child/Adolescent

- | | |
|--|--|
| 1. Construction Task | 10. Socioemotional Questions:
Social Difficulties and
Annoyance |
| 2. Make-Believe Play | 11. Break |
| 3. Joint Interactive Play | 12. Socioemotional Questions:
Friends, Relationships, and
Marriage |
| 4. Demonstration Task | 13. Socioemotional Questions:
Loneliness |
| 5. Description of a Picture | 14. Creating a Story |
| 6. Telling a Story From a Book | |
| 7. Cartoons | |
| 8. Conversation and Reporting | |
| 9. Socioemotional Questions:
Emotions | |

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Construction Task



The image shows a book titled "ANALYSIS of Construction from Task Materials (2, 3, and 4)" by W. P. K. The book is open, and the left page displays a 3D geometric construction made of blue and pink chevron-shaped blocks. The construction is a symmetrical, stepped pyramid-like shape. The right page is blank. The book is resting on a white surface.

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Make-Believe Play and Joint Interactive Play



A collection of various toys arranged on a white background. The toys include a white measuring cup with red markings, a red fire truck, a white CD, a red basketball, a stack of white blocks on a purple base, a white spoon and fork, a small white dinosaur, a knight in white armor, and a superhero figure in a blue and red suit. There are also some small white plates and a small white bottle.

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Construction Task

Modules 2, 3, and 4 Administration

1. Place puzzle pieces on the table so the participant can see them but cannot reach them.
2. Place the printed design in front of the participant and instruct them to fill in the picture with all of the pieces and to let you know when they need more.
3. Wait for the participant to request more pieces before giving them. If they do not make the request, deliberately look at them.
4. If they do not respond to your gaze, check in with them verbally, and if they do not respond, verbally offer more pieces.
5. Continue the task until all the pieces are in place.
6. When finished, place the container in front of the participant and open it, while verbally prompting a transition and starting to put pieces in.
7. Observe whether the participant helps in putting pieces away or not.

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Make-Believe Play and Joint Interactive Play

Modules 2 and 3 Administration

1. Introduce action figures with descriptions.
2. Encourage participant to play with the toys—suggest making up a story or video game.
3. If participant does not begin to play, model play with some objects. Verbally inquire how they are going to play with their toys.
4. Comment, show interest, and encourage the participant without telling them what to do.
5. It is helpful to ask them what they are doing throughout the activity.
6. After a few minutes, ask if you can play, too.
7. Join them in play; observe whether they can follow your ideas as well as suggest their own ideas.

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Demonstration Task



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Demonstration Task Modules 2, 3, and 4 Administration

1. Initiate the task: "Let's pretend that I don't know how to brush my teeth, and I need you to teach me!"
2. Set the scene with imaginary gestures, being sure to avoid acting out any tasks
 - Sink, hot water, cold water, toothbrush, toothpaste, and the cup
3. Reiterate your desire for the participant to show and tell you how to brush your teeth and to start at the beginning upon entry into the bathroom.
4. If participant only uses gestures, have them restart, emphasizing show and tell. Do the same if participant speaks instructions without gestures.
5. If you observe that the participant doesn't understand the task, reiterate instructions and/or demonstrate an unrelated event (driving a car).
6. Then re-draw scene, instructing participant to show and tell you how they wash hands (which is simpler than tooth brushing).
7. If participant still does not show and tell, re-draw scene using real props.

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Description of a Picture



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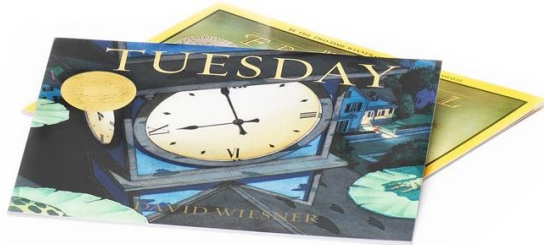
Description of a Picture Modules 2, 3, and 4 Administration

1. Instruct the participant to look at the picture, followed by "What is happening here?"
2. Allow time for the participant to comment spontaneously. If they do not, provide only isolated specific labels of objects or people.
3. Provide encouragement and amplification of the participant's answers.
4. Present the second picture only if the participant is not interested in the first picture and does not respond.

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Telling a Story From a Book



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Telling a Story From a Book Modules 2, 3, and 4 Administration

1. Present one of the books to the participant and talk about how you are going to look at the book, how it tells a story, and how you will both take turns telling the story.
2. Ask the participant to look at it and tell you the story.
3. Hand the book to the participant and provide whatever prompts necessary to encourage them to begin describing the story and then to turn the page.
4. Do not model labeling emotions.
5. Allow the participant a few minutes to describe the book.
6. Try to take a turn part way through the book, but allow them to finish the book if they insist.

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Cartoons



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Cartoons Modules 3 and 4 Administration

1. Tell the participant they will be shown cartoons, then asked to retell the story without looking at the pictures.
2. Look through the cards one by one with the participant, encouraging commentary.
3. Ask the participant to move away from the table, stand up, and tell the story (it can be helpful if you stand up, too).
4. It is important the participant does not have anything in their hands and is not touching any surfaces.
5. Proceed to the second series of cartoons if the first is "spoiled" (e.g., participant insists on looking at cards during the first cartoon re-telling).

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Conversation, Reporting, and Interview Questions

- Conversation and Reporting
- Socioemotional Questions: Emotions
- Socioemotional Questions: Social Difficulties and Annoyance

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Break



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Break Modules 3 and 4 Administration

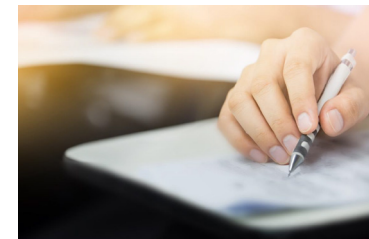
1. Say, "Let's take a break," and indicate that you need some time to catch up on your notes.
2. Point to the "Break" materials and express the hope that the participant can find something of interest to them.
3. Say, "I need to make some notes for a few minutes," and move your chair back from the table so you are sitting within view but away from the table.
4. Note the participant's interest in materials, and work on your notes for at least 2 minutes.
5. Observe whether the participant initiates an interaction. If so, respond briefly and positively, but indicate you need to finish your paperwork.
6. After a few seconds look up, catch the participant's eye, and smile in encouragement. The goal is to create an occasion for the participant to initiate an interaction. If this does not occur, you can return to your notes or say, "I'll just be another minute."
7. Return to the table after a few minutes, making it clear that you are finished taking notes.

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Interview Questions

- Socioemotional Questions: Friends, relationships, and marriage
- Socioemotional Questions: Loneliness



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General Guidelines About Interview Questions

- Proceed through all of the questions in order to ensure that the information necessary for coding is elicited.
- Although minor wording changes may be necessary (e.g., when adapting for a young child or adult who is already married), be sure to maintain the “spirit” of the questions and ask each question in order to give every participant the same opportunities.
- The questions are not meant to replace or mimic a psychiatric interview; you will need additional information outside of the ADOS-2 to assess for co-occurring symptoms and diagnoses.

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Creating a Story



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Creating a Story Modules 3 and 4 Administration

- Tell the participant that you both are going to make up stories using the objects you have, that you will go first, and then it will be their turn.
- Model choosing 5 objects and making up a simple story that is at the appropriate level for the participant.
 - Use 3 in a creative way and 2 as the actual object (e.g., umbrella as an umbrella).
- Place your objects to the side when you are finished.
- Encourage the participant to pick 5 new objects to use in making up their own story.

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Read Through the Coding Sections

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List of Codes: Module 3

A Codes: Language and Communication

Overall Level of Non-Echoed Spoken Language

Speech Abnormalities Associated With Autism

Immediate Echolalia

Stereotyped/Idiosyncratic Use of Words or Phrases

Offers Information

Asks for Information

Reporting of Events

Conversation

Descriptive, Conventional, Instrumental or Informational Gestures

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List of Codes: Module 3 (cont.)

B Codes: Reciprocal Social Interaction

Unusual Eye Contact

Facial Expressions Directed to Others

Language Production and Linked Nonverbal Communication

Shared Enjoyment in Interaction

Comments on Others' Emotions/Empathy

Insight Into Typical Social Situations and Relationships

Quality of Social Overtures

Amount of Social Overtures/Maintenance of Attention

Quality of Social Response

Amount of Reciprocal Social Communication

Overall Quality of Rapport

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List of Codes: Module 3 (cont.)

C Code: Imagination	D Codes: Stereotyped Behaviors and Restricted Interests	E Codes: Other Abnormal Behaviors
Imagination/Creativity	Unusual Sensory Interest in Play Materials/Person	Overactivity/Agitation
	Hand and Finger and Other Complex Mannerisms	Tantrums, Aggression, Negative or Disruptive Behavior
	Self-Injurious Behavior	Anxiety
	Excessive Interest in or References to Unusual or Highly Specific Topics or Objects or Repetitive Behaviors	
	Compulsions or Rituals	

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ADOS-2 Administration and Independent Scoring

Module 3 or 4

Algorithm

- Transfer selected scores from protocol to algorithm
- Convert scores
 - 3 to 2
 - 7, 8, and 9 to 0

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Classification: Module 3

- Compute algorithm domain total scores for Social Affect (SA) and Restricted and Repetitive Behavior (RRB)
- Combine SA + RRB totals = Overall Total
- Compare the Overall Total to the cutoff scores provided for each algorithm
- This will result in an ADOS-2 Classification, NOT a diagnosis*
 - An individual's ADOS-2 Classification may or may not be the same as their clinical diagnosis

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Classification: Module 3 (cont.)

Compare the Overall Total to the cutoff scores....

- Fill in ADOS-2 Classification
 - Classification, NOT a diagnosis

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Comparison Score: Module 3

- Use Overall Total score and chronological age to compute Comparison Score
- Comparison Score table is provided in Manual and on protocol

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Comparison Score: Module 3 (cont.)



Convert Overall Total Score to Comparison Score....

ADOS-2 Module 3 Algorithm

Participant ID: _____ Examiner: _____
 Session Report (Module 3): _____ Date of Evaluation: _____ Chronological Age: _____

CONVERTING TO ALGORITHM SCORES
 *Follow the algorithm scores of the algorithm scores of 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 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1508, 1509, 1510, 1511, 1512, 1513, 1514, 1515, 1516, 1517, 1518, 1519, 1520, 1521, 1522, 1523, 1524, 1525, 1526, 1527, 1528, 1529, 1530, 1531, 1532, 1533, 1534, 1535, 1536, 1537, 1538, 1539, 1540, 1541, 1542, 1543, 1544, 1545, 1546, 1547, 1548, 1549, 1550, 1551, 1552, 1553, 1554, 1555, 1556, 1557, 1558, 1559, 1560, 1561, 1562, 1563, 1564, 1565, 1566, 1567, 1568, 1569, 1570, 1571, 1572, 1573, 1574, 1575, 1576, 1577, 1578, 1579, 1580, 1581, 1582, 1583, 1584, 1585, 1586, 1587, 1588, 1589, 1590, 1591, 1592, 1593, 1594, 1595, 1596, 1597, 1598, 1599, 1600, 1601, 1602, 1603, 1604, 1605, 1606, 1607, 1608, 1609, 1610, 1611, 1612, 1613, 1614, 1615, 1616, 1617, 1618, 1619, 1620, 1621, 1622, 1623, 1624, 1625, 1626, 1627, 1628, 1629, 1630, 1631, 1632, 1633, 1634, 1635, 1636, 1637, 1638, 1639, 1640, 1641, 1642, 1643, 1644, 1645, 1646, 1647, 1648, 1649, 1650, 1651, 1652, 1653, 1654, 1655, 1656, 1657, 1658, 1659, 1660, 1661, 1662, 1663, 1664, 1665, 1666, 1667, 1668, 1669, 1670, 1671, 1672, 1673, 1674, 1675, 1676, 1677, 1678, 1679, 1680, 1681, 1682, 1683, 1684, 1685, 1686, 1687, 1688, 1689, 1690, 1691, 1692, 1693, 1694, 1695, 1696, 1697, 1698, 1699, 1700, 1701, 1702, 1703, 1704, 1705, 1706, 1707, 1708, 1709, 1710, 1711, 1712, 1713, 1714, 1715, 1716, 1717, 1718, 1719, 1720, 1721, 1722, 1723, 1724, 1725, 1726, 1727, 1728, 1729, 1730, 1731, 1732, 1733, 1734, 1735, 1736, 1737, 1738, 1739, 1740, 1741, 1742, 1743, 1744, 1745, 1746, 1747, 1748, 1749, 1750, 1751, 1752, 1753, 1754, 1755, 1756, 1757, 1758, 1759, 1760, 1761, 1762, 1763, 1764, 1765, 1766, 1767, 1768, 1769, 1770, 1771, 1772, 1773, 1774, 1775, 1776, 1777, 1778, 1779, 1780, 1781, 1782, 1783, 1784, 1785, 1786, 1787, 1788, 1789, 1790, 1791, 1792, 1793, 1794, 1795, 1796, 1797, 1798, 1799, 1800, 1801, 1802, 1803, 1804, 1805, 1806, 1807, 1808, 1809, 1810, 1811, 1812, 1813, 1814, 1815, 1816, 1817, 1818, 1819, 1820, 1821, 1822, 1823, 1824, 1825, 1826, 1827, 1828, 1829, 1830, 1831, 1832, 1833, 1834, 1835, 1836, 1837, 1838, 1839, 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2172, 2173, 2174, 2175, 2176, 2177, 2178, 2179, 2180, 2181, 2182, 2183, 2184, 2185, 2186, 2187, 2188, 2189, 2190, 2191, 2192, 2193, 2194, 2195, 2196, 2197, 2198, 2199, 2200, 2201, 2202, 2203, 220

List of Codes Module 4

A Codes: Language and Communication	B Codes: Reciprocal Social Interaction
Overall Level of Non-Echoed Spoken Language	Unusual Eye Contact
Speech Abnormalities Associated With Autism	Facial Expressions Directed to Others
Immediate Echolalia	Language Production and Linked Nonverbal Communication
Stereotyped/Idiosyncratic Use of Words or Phrases	Shared Enjoyment in Interaction
Offers Information	Communication of Own Affect
Asks for Information	Comments on Others' Emotions/Empathy
Reporting of Events	Insight Into Typical Social Situations and Relationships
Conversation	Responsibility
Descriptive, Conventional, Instrumental, or Informational Gestures	Quality of Social Overtures
Emphatic or Emotional Gestures	Amount of Social Overtures/Maintenance of Attention
	Quality of Social Response
	Amount of Reciprocal Social Communication
	Overall Quality of Rapport

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List of Codes Module 4 (cont.)

C Code: Imagination	D Codes: Stereotyped Behaviors and Restricted Interests	E Codes: Other Abnormal Behaviors
Imagination/Creativity	Unusual Sensory Interest in Play Material/Person	Overactivity/Agitation
	Hand and Finger and Other Complex Mannerisms	Tantrums, Aggression, Negative or Disruptive Behavior
	Self-Injurious Behavior	Anxiety
	Excessive Interest in or References to Unusual or Highly Specific Topics or Objects or Repetitive Behaviors	
	Compulsions or Rituals	

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Classification: Module 4 (Original Algorithm)

Converting the Communication Total, Social Interaction Total, and Communication + Social Interaction Total to the ADOS-2 Classification

	COMMUNICATION	SOCIAL INTERACTION	COMMUNICATION + SOCIAL INTERACTION
autism	3	6	10
autism spectrum	2	4	7

Assign the ADOS-2 Classification:

autism	All three totals (Communication Total, Social Interaction Total, combined Communication + Social Interaction Total) are equal to or greater than the three separate corresponding autism cutoffs: • All of the following are true: Communication Total is 3 or higher AND Social Interaction Total is 6 or higher AND Communication + Social Interaction Total is 10 or higher
autism spectrum	All three totals (Communication Total, Social Interaction Total, combined Communication + Social Interaction Total) are equal to or greater than the three separate corresponding autism spectrum cutoffs, but at least one is less than its corresponding autism cutoff: • All of the following are true: Communication Total is 2 or higher AND Social Interaction Total is 4 or higher AND Communication + Social Interaction Total is 7 or higher AND • At least one of the following is also true: Communication Total is 2; Social Interaction Total is 4 or 5; Communication + Social Interaction Total is 7 to 9
non-spectrum	Any one of the three totals (Communication Total, Social Interaction Total, combined Communication + Social Interaction Total) is less than the autism spectrum cutoff: • At least one of the following is true: Communication Total is 1 or lower; Social Interaction Total is 3 or lower; Communication + Social Interaction Total is 6 or lower

Record the ADOS-2 Classification on the front of this form in the space marked ADOS-2 Classification.

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Classification: Module 4 (Original Algorithm) (cont.)

Example:

Met autism spectrum cutoff [2] on Communication Total,

Exceeded autism cutoff [8] on Social Interaction Total,

Met autism cutoff [10] on COM + SI Total

Classification = **autism spectrum**

Example:

Met autism spectrum cutoff [2] on Communication (COM) Total,

Met autism spectrum cutoff [4] on Social Interaction (SI) Total,

Did not meet autism spectrum cutoff [6] on COM + SI Total

Classification = **non-spectrum**

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Classification: Module 4 (Original Algorithm) (cont.)



- For ADOS-2 Classifications of autism spectrum and autism, must meet cutoffs in *all three areas*:
 - Communication Total
 - Social Interaction Total
 - Communication + Social Interaction Total
- Imagination/Creativity and Stereotyped Behaviors and Restricted Interests are not included in algorithm totals.

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Revised Algorithm for Module 4

Language and Communication

Conversation (A-8) _____
Emphatic or Emotional Gestures (A-10) _____

Reciprocal Social Interaction

Unusual Eye Contact (B-1) _____
Facial Expressions Directed to Others (B-2) _____
Communication of Own Affect (B-5) _____
Insight (B-7) _____
Quality of Social Overtures (B-9) _____
Quality of Social Response (B-11) _____
Amount of Reciprocal Social Communication (B-12) _____
Overall Quality of Rapport (B-13) _____

Social Affect (SA) Total _____

Restricted and Repetitive Behaviors

Speech Abnormalities Associated with Autism (A-2) _____
Stereotyped/Idiosyncratic Use of Words or Phrases (A-4) _____
Unusual Sensory Interest in Play Material/Person (D-1) _____
Hand and Finger and Other Complex Mannerisms (D-2) _____
Excessive Interest in or References to Highly Specific Topics (D-4) _____

Restricted and Repetitive Behavior (RRB) Total _____

SA and RRB Total _____ (autism spectrum cutoff = 8)

Source: Hus, V., & Lord, C. (2014). The ADOS Module 4: Revised algorithm and standardized severity scores. *Journal of Autism and Developmental Disorders*, 44(8), 1996–2012.

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Revised Module 4 Algorithm Calibrated Severity Scores (i.e., Comparison Scores)

- SA domain total + RRB domain total = Overall Total Score
 - ASD cutoff= 8 on Overall Total Score
- Find corresponding CSS from each total score (Overall, SA domain, RRB domain)

CSS	Raw total		
	Overall Total	SA domain	RRB domain
1	0-2	0-1	0
2	3-5	2-3	-
3	6-7	4	-
4	8	5	-
5	9	6	1
6	10-11	7-8	2
7	12-13	9-10	3
8	14-15	11-12	4
9	16-19	13-15	5
10	20-29	16-20	6-10

Source: Hus, V., & Lord, C. (2014). The ADOS Module 4: Revised algorithm and standardized severity scores. *Journal of Autism and Developmental Disorders*, 44(8), 1996–2012.

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Calculating Reliability

- Goal: **At least 80%** interrater agreement or agreement with consensus on overall protocol and algorithm items
- Protocol Reliability = $\frac{\text{Number Agree}}{\text{Total Number of Items}^*}$
- Algorithm Reliability = $\frac{\text{Number Agree on Algorithm}}{\text{Total Number of Algorithm Items}^*}$

*Do **not include** items consensus-coded as 8 or 9 in numerator or denominator, unless you coded differently.

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Protocol Reliability

- Step 1: Add coding sections to the front of the protocol.
- Step 2: After each coding section is completed, count all the items where you agreed with consensus coding and add to the front of your protocol.
- Step 3: Add all items you had in agreement between all coding sections; this will be your numerator.
- Step 4: Put your numerator over the number of items in the protocol (example: $17/28 = 60\%$).

*29 items total, but use 28 because of the 8 on B-3

ADOS-2 **Module 3 Algorithm**

Participant ID: _____ Examiner: _____
 Gender: Female ☐ Male ☐ Date of Birth: _____ Date of Evaluation: _____
 Chronological Age: _____

CONVERTING ITEM CODES TO ALGORITHM SCORES

- Convert assigned ratings of 3 to algorithm scores of 2.
- Convert assigned ratings other than 0, 1, 2, or 3 (A, 7, 8, and 9) to algorithm scores of 0.
- Transfer assigned ratings of 0, 1, and 2 directly to the algorithm form (do not convert).

Social Affect (SA)

Communication			
Reporting of Events	(A-7)	1	0
Conversation	(A-8)	1	2
Descriptive, Conversational, Instrumental, or Informational	(A-9)	2	2
Reciprocal Social Interaction	(B-1)	2	1
Unusual Eye Contact	(B-2)	1	0
Facial Expressions Directed to	(B-3)	1	2
Shared Enjoyment in	(B-4)	1	0
Quality of Social Overlap	(B-7)	1	2
Quality of Social Response	(B-9)	1	0
Amount of Reciprocal Social Communication	(B-10)	0	2
Overall Quality of Rapport	(B-11)	1	2
SA TOTAL		11	

Restricted and Repetitive Behavior (RRB)

Restricted and Repetitive Behaviors			
Stereotyped/Idiosyncratic Use of Words or Phrases	(A-4)	2	2
Unusual Sensory Interest in Play Material/Person	(D-1)	2	2
Hand and Finger and Other Complex Mannerisms	(D-2)	2	1
Excessive Interest in Unusual or Highly Specific Topics/Objects or Repetitive Behaviors	(D-4)	2	2
RRB TOTAL		6	
OVERALL TOTAL (SA + RRB)		17	

A Codes: 5
 B Codes: 5
 C Code: 0
 D Codes: 4
 E Codes: 3

17 correct
 *28 total items

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Algorithm Reliability

ADOS-2 **Module 3 Algorithm**

Participant ID: _____ Examiner: _____
 Gender: Female ☐ Male ☐ Date of Birth: _____ Date of Evaluation: _____
 Chronological Age: _____

CONVERTING ITEM CODES TO ALGORITHM SCORES

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Conversation	(A-8)	1	2
Descriptive, Conversational, Instrumental, or Informational	(A-9)	2	2
Reciprocal Social Interaction	(B-1)	2	1
Unusual Eye Contact	(B-2)	1	0
Facial Expressions Directed to	(B-3)	1	2
Shared Enjoyment in	(B-4)	1	0
Quality of Social Overlap	(B-7)	1	2
Quality of Social Response	(B-9)	1	0
Amount of Reciprocal Social Communication	(B-10)	0	2
Overall Quality of Rapport	(B-11)	1	2
SA TOTAL		11	

Restricted and Repetitive Behavior (RRB)

Restricted and Repetitive Behaviors			
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Excessive Interest in Unusual or Highly Specific Topics/Objects or Repetitive Behaviors	(D-4)	2	2
RRB TOTAL		6	
OVERALL TOTAL (SA + RRB)		17	

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Make a column next to your codes and write in trainer's codes

Algorithm Reliability (cont.)

ADOS-2 **Module 3 Algorithm**

Participant ID: _____ Examiner: _____
 Gender: Female ☐ Male ☐ Date of Birth: _____ Date of Evaluation: _____
 Chronological Age: _____

CONVERTING ITEM CODES TO ALGORITHM SCORES

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Conversation	(A-8)	1	2
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Reciprocal Social Interaction	(B-1)	2	1
Unusual Eye Contact	(B-2)	1	0
Facial Expressions Directed to	(B-3)	1	2
Shared Enjoyment in	(B-4)	1	0
Quality of Social Overlap	(B-7)	1	2
Quality of Social Response	(B-9)	1	0
Amount of Reciprocal Social Communication	(B-10)	0	2
Overall Quality of Rapport	(B-11)	1	2
SA TOTAL		11	

Restricted and Repetitive Behavior (RRB)

Restricted and Repetitive Behaviors			
Stereotyped/Idiosyncratic Use of Words or Phrases	(A-4)	2	2
Unusual Sensory Interest in Play Material/Person	(D-1)	2	2
Hand and Finger and Other Complex Mannerisms	(D-2)	2	1
Excessive Interest in Unusual or Highly Specific Topics/Objects or Repetitive Behaviors	(D-4)	2	2
RRB TOTAL		6	
OVERALL TOTAL (SA + RRB)		17	

Circle your codes that are the same as the trainer's codes.

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Algorithm Reliability (cont.)

ADOS-2 **Module 3 Algorithm**

Participant ID: _____ Examiner: _____
 Gender: Female ☐ Male ☐ Date of Birth: _____ Date of Evaluation: _____
 Chronological Age: _____

CONVERTING ITEM CODES TO ALGORITHM SCORES

- Convert assigned ratings of 3 to algorithm scores of 2.
- Convert assigned ratings other than 0, 1, 2, or 3 (A, 7, 8, and 9) to algorithm scores of 0.
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Social Affect (SA)

Communication			
Reporting of Events	(A-7)	1	0
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Descriptive, Conversational, Instrumental, or Informational	(A-9)	2	2
Reciprocal Social Interaction	(B-1)	2	1
Unusual Eye Contact	(B-2)	1	0
Facial Expressions Directed to	(B-3)	1	2
Shared Enjoyment in	(B-4)	1	0
Quality of Social Overlap	(B-7)	1	2
Quality of Social Response	(B-9)	1	0
Amount of Reciprocal Social Communication	(B-10)	0	2
Overall Quality of Rapport	(B-11)	1	2
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Restricted and Repetitive Behavior (RRB)

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Hand and Finger and Other Complex Mannerisms	(D-2)	2	1
Excessive Interest in Unusual or Highly Specific Topics/Objects or Repetitive Behaviors	(D-4)	2	2
RRB TOTAL		6	
OVERALL TOTAL (SA + RRB)		17	

Step 1:
Count up the number of codes you got correct (example = 7 correct).

Step 2:
Divide the number of correct items by 14 to obtain percent correct/reliability (example: $7/14 = 50\%$).

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Questions?



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Day 2 Overview

- Day 2 starts at 8:30 a.m.
- Introduction to Modules 1 and 2
- Demonstration of Module 1 or 2
- Coding
- ADOS-2 Psychometrics
- Appropriate use of the ADOS-2
- Information for further preparation and training

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